

# Mental Imagery and the Third Dimension

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## SUMMARY

What sort of medium underlies imagery for three-dimensional scenes? In the present investigation, the time subjects took to scan between objects in a mental image was used to infer the sorts of geometric information that images preserve. Subjects studied an open box in which five objects were suspended, and learned to imagine this display with their eyes closed. In the first experiment, subjects scanned by tracking an imaginary point moving in a straight line between the imagined objects. Scanning times increased linearly with increasing distance between objects in three dimensions. Therefore metric 3-D information must be preserved in images, and images cannot simply be 2-D "snapshots." In a second experiment, subjects scanned across the image by "sighting" objects through an imaginary rifle sight. Here scanning times were found to increase linearly with the two-dimensional separations between objects as they appeared from the original viewing angle. Therefore metric 2-D distance information in the original perspective view must be preserved in images, and images cannot simply be 3-D "scale models" that are accessed from any and all directions at once. In a third experiment, subjects mentally rotated the display 90° and scanned between objects as they appeared in this new perspective view by tracking an imaginary rifle sight, as before. Scanning times increased linearly with the two-dimensional separations between objects as they would appear from the new relative viewing perspective. Therefore images can display metric 2-D distance information in a perspective view never actually experienced, so mental images cannot simply be "snapshot plus scale model" pairs.

These results can be explained by a model in which the three-dimensional structure of objects is encoded in long-term memory in 3-D object-centered coordinate systems. When these objects are imagined, this information is then mapped onto a single 2-D "surface display" in which the perspective properties specific to a given viewing angle can be depicted.

In a set of perceptual control experiments, subjects scanned a visible display by (a) simply moving their eyes from one object to another, (b) sweeping an imaginary rifle sight over the display, or (c) tracking an imaginary point moving from one object to another. Eye-movement times varied linearly with 2-D interobject distance, as did time to scan with an imaginary rifle sight; time to track a point varied independently with the 3-D and 2-D interobject distances. These results are compared with the analogous image scanning results to argue that imagery and perception share some representational structures but that mental image scanning is a process distinct from eye movements or eye-movement commands.

How do people mentally represent physical space? Attneave (1972, 1974) has proposed a rather straightforward answer to this question: Three-dimensional physical space is represented in an internal three-dimensional "space" or coordinate system. Physical ob-

jects are represented in that "space" by "filled-in" regions with the same shape as the object, like scale models. This "sandbox in the head" theory, as Attneave calls it, was motivated by the ease and accuracy with which people can mentally perform

smooth spatial transformations in imagined scenes. For example, Shepard and Metzler (1971) showed that people can mentally rotate an image of one three-dimensional object to bring it into correspondence with a second object depicted at a different orientation. In doing so, subjects required proportionally more time to rotate the image greater amounts, and took approximately the same amount of time whether they rotated the image in the picture plane or in depth. In the same vein, Pinker and Kosslyn (1978; see also Pinker, 1979) showed that when people mentally scan in a straight line between two objects in an imagined three-dimensional scene, they require proportionally more time to scan between objects separated by greater distances in three dimensions. Finally, Attneave and Pierce (1978) have shown that when people mentally extrapolate a visible pointer through space, they are equally accurate when extrapolating it through visible space and through the imagined space behind their heads.

Thus, the argument goes, since people can perform mental analogues of rotation in depth and tracing a straight line through space, there must be some internal three-dimensional medium in which the rotation or scanning takes place (Attneave, 1972, 1974; Metzler & Shepard, 1974; Pinker & Kosslyn, 1978). In this view, it is the representation of the scene in the three-dimensional (3-D) medium, and not some two-dimensional or photographic representation, that

is processed during the scanning or rotation of images.

The claim that mental imagery involves direct processing of three-dimensional model-like structures has important and somewhat surprising consequences. These consequences can be most easily seen by contrasting a concrete example of a system lacking direct access to the three-dimensional structure of objects—the visual sense—with a concrete example of a system with such direct access—the haptic sense. In vision, the three-dimensional layout of a scene is not processed directly, but is inferred or reconstructed from the two-dimensional projections of the scene onto the retinal surfaces. As a consequence of the laws of projective geometry, *perspective effects*, which depend on the angle and distance of the viewer, exist in vision. For example, an object will subtend a smaller visual angle when it recedes from the viewer, will foreshorten as it is rotated in depth, and will be occluded if a nearer object interrupts the line of sight. In contrast, when a scene is explored by touch, its 3-D structure is experienced directly, and as a consequence, there are no accompanying perspective effects. More distant objects do not feel smaller, nor does a tilted rectangle feel like a trapezoid, nor is the back of an object inaccessible to the touch. Thus, a straightforward interpretation of the sandbox theory entails that imagery for three-dimensional scenes should resemble touch more than vision, notwithstanding the widespread consensus that imagery and vision are governed by similar principles. Consider the alternative: Surely no one would suggest that three-dimensional images reflect some sort of mental “light rays,” nor that the mind’s eye contains a lens and a retina onto which “images” of images are projected! And the inescapable consequence of this direct access to 3-D structure is that people should not experience perspective effects when they imagine scenes, just as they do not experience these effects when they explore a scene with their hands.

Thus, it is somewhat of a paradox that there should exist evidence that perspective properties *are* experienced in imagery, as if the images *were* being “seen” from a particular “vantage point.” For example,

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Attneave and Pierce (1978) reported that most subjects claimed to have been unable to imagine the scene in front of them and behind them simultaneously, and Neisser and Kerr (1973) found that most subjects could not simultaneously be aware of an imagined object and "conceal" the object inside another one. Several kinds of empirical results support these introspective reports. For example, when people are asked to imagine a scene from a particular vantage point, they are actually less likely to remember details of the scene that were not "visible" from the imagined "viewing perspective" (Abelson, 1976; Fiske, Taylor, Etcoff, & Laufer, 1979; Keenan & Moore, 1979). In addition, Kosslyn (1978) has found that the "visual angle" subtended by an imagined object seems to increase linearly with how "near" the object appears in the image. Finally, the fact that people must mentally rotate an object into correspondence with a second one to determine that both have the same shape (Shepard & Metzler, 1971) suggests that the mental representation of those objects preserves some of the information associated with seeing the objects from a particular perspective—otherwise the two objects could be matched directly, without rotation (as Metzler & Shepard, 1974, point out).<sup>1</sup>

In sum, it is difficult to see how images could be directly accessible 3-D structures and exhibit perspective properties. However, the evidence at present is only suggestive and far too sketchy to support the claim that a genuine paradox is at hand. We simply lack systematic evidence about the sorts of information, three-dimensional or perspective-specific, that are preserved in mental images. The present investigation is an attempt to measure these sorts of information, with the goal of further specifying the nature of the medium that underlies mental images for three-dimensional scenes. In particular, the following questions are raised: (a) Do mental images preserve interval information concerning the distances between objects in three dimensions? (b) Do mental images preserve interval information about the two-dimensional or projected distances between objects as they appeared from the original angle of view of

the scene? (c) Can mental images be used to display two-dimensional interobject distances as they would appear from a new vantage point never directly experienced? (d) What is the relationship between the representational medium used in imagery for three-dimensional scenes and the one used in the perception of those scenes? We attempt to answer these questions using an image-scanning paradigm similar to the one used by Kosslyn, Ball, and Reiser (1978) and Pinker and Kosslyn (1978). In this method, the time that subjects require to scan from one object in an image to another is taken as a measure of the "distance" between those objects in the image.

### Experiment 1

Pinker and Kosslyn (1978) found that scanning times increased with increasing distance in three dimensions between objects in a memorized stimulus display. However, these correlations were not high enough to demonstrate with certainty that interval information was preserved in the image. If the conditions in that experiment corresponding to no scanning at all are deleted (i.e., the instruction to scan "from the car to the car"), none of the correlations found exceeded .80. Thus, it is possible that subjects were unable to encode precise locations of objects in three-dimensional space, but simply remembered whether they were near, far, or an intermediate distance away

<sup>1</sup> This fact does not necessarily imply that perspective properties specific to the original angle of view are preserved, however. Let us assume that the Shepard and Metzler subjects encoded each object as a set of points or lines within a three-dimensional coordinate system. Let us assume further that the axes of the coordinate system were always defined in the same way relative to the viewer, say, with one axis coinciding with the line of sight, the second with the gravitational vertical, and the third parallel to the horizon. Finally, let us assume that the same—different judgment is made by matching one object representation against the other in a template fashion. Clearly, one object's representation must be brought into the same orientation relative to its coordinate system as the other in order for the template match to yield the desired result. If this normalization process occurred incrementally, response times would vary with angular disparity. On this account, the viewer specificity would be preserved in the encoding of 3-D shape relative to the axes and not in a 2-D depiction of the perspective view.

(see also Pinker, 1979, for further analyses of the Pinker and Kosslyn data). I decided, then, to begin by replicating part of the Pinker and Kosslyn experiment, using more subjects and trials for each condition, in order to obtain more stable data. In this experiment I asked subjects to scan in straight lines (by imagining a moving point) between every possible pair of objects (excluding pairs consisting of an object with itself). These scanning times are used as a kind of "tape measure," allowing one to discern whether the 3-D distances were in fact preserved in the image. If so, then scan times should be highly correlated with these distances and not significantly correlated with other measures of interobject distance.

### Method

#### Subjects

Eight undergraduates, one graduate student, and one research assistant, all affiliated with Harvard University, volunteered to be subjects in this experiment. Subjects participating in this and all of the subsequent experiments reported in this article were paid for their time and were not familiar with the hypotheses under investigation.

#### Materials

**Visual stimuli.** A  $38 \times 38$  cm light gray box, open at the top and front, was located 51 cm away from a chinrest positioned so that subjects were looking into the center of the box. Five small toys (each less than 5 cm long), a hat, an apple, a teddy bear, a tire, and a sea shell, were suspended by clear nylon thread from flat wooden sticks ( $79 \times 1.25$  cm) lying across the top of the box parallel to the front edge. A 3-mm green dot was affixed to the center of each object. The objects' positions were chosen so that the interobject distances in three dimensions correlated poorly (.29) with the corresponding distances in the two-dimensional parallel projection of the objects' positions onto the frontal plane.

**Trials.** A trial consisted of the naming of an object (the "source" object), a 4-sec silence, and the naming of a second object (the "destination" object). A new source object was named 4 sec after the subject responded, beginning a new trial.

Blocks of 15 trials were constructed, each containing the 10 possible pairs of objects plus 5 additional trials that paired each object with an object that was not in the box (pig, car, face, shoe, tree). The trials were randomly ordered within a block under the constraints that no destination object could appear in the immediately preceding trial, that no object could be mentioned in three consecutive trials, and that neither type of trial (destination object present or absent) could appear in

more than three consecutive trials. Seven such blocks were constructed: The first was a practice block, the data from which (unbeknown to the subject) were ignored; the next six blocks were coupled so that the order of source and destination objects within each trial was counterbalanced across the successive pairs of blocks.

Trials were tape recorded and replayed on a two-channel relay-controlled tape recorder. For each trial both members of the object pair were recorded on one channel, which was played to the subject. Only the second member was recorded on the second channel; its onset started a digital millisecond clock and stopped the tape recorder after a .6-sec delay. The subjects' pressing either of two telegraph keys stopped the clock and restarted the tape recorder; this arrangement assured a constant 4-sec intertrial interval.

#### Procedure

Subjects, tested individually, were told that they were participating in an experiment on visual memory and were asked to study the scene in front of them with chin in chinrest. They were asked to form a mental image of the box and its contents, making sure each object was imagined at its proper location. After the subject claimed to be able to form an accurate image, the experimenter singled out an object, gave the subject an opportunity to study its position, removed the object from the box, and asked the subject to tell him how to replace the object in its former location. The subject was to use directions like "place the object roughly over there, then slide it to the right until I say 'stop,' then slide it back until I say 'stop,' " and so on. The experimenter moved the stick from which the object hung in accordance with these directions, and when the subject was satisfied, the experimenter measured the accuracy of the placement. This procedure was repeated until the subject could direct placement of the object to within 1.25 cm of its original position; then the entire procedure was repeated for each of the other four objects. The experimenter then randomly rearranged the five objects in the box and asked the subject to direct all five back to their original positions; this too was repeated until all objects were repositioned with sufficient accuracy. (If the subject correctly positioned four out of five objects on one of those trials, he or she was required to study and reposition only the inaccurate object, rather than all five.) It took subjects from one to three attempts to position a single object and from two to five attempts to position all of them at once. The experimenter covered the front of the box with an opaque screen at the conclusion of this training phase of the experiment.

After memorizing the layout of the stimulus scene, the subject was asked to place his or her hands on each of the two telegraph keys in front of him or her; the key under the dominant hand was labelled *true*; the other one, *false*. The subject was asked to shut his or her eyes, and to listen to the tape. Upon hearing a name, the subject was to form a mental image of the box and its contents and to "focus on" the object that was named. The subject was asked to hold the image and remain focused on this object until the next object

was named. If the second object named was in the box, the subject was to "scan" to it by focusing on a point or a small black dot moving smoothly in a straight line as quickly as possible from the first to the second object. It was stressed that the subject should "see" the point at all times as it moved along its path, to assure that he or she would, in fact, scan the entire straight path between objects. When the subject "arrived" at the destination object, he or she was to press the *true* key. On those trials in which the second object named was not in the box, the subject was to consult his or her image to be sure that it was not there and then to press the *false* key.<sup>2</sup> The subjects were asked to perform the task at the fastest possible rate while still following all the instructions and responding as accurately as possible.

Prior to starting the tape, the experimenter gave the subject 4–5 untimed practice trials and asked whether he or she was experiencing any difficulty in following the instructions; if so, 4–5 additional practice trials were given. After a final review of the instructions, the experimenter started the tape recorder. A short break followed the end of the fourth block of trials; the boundaries between blocks were not designated in any other way. When the tape was over, the subject was asked to fill out a form containing the following four questions: "In what percentage of the trials did you follow the instructions to scan an image?" "If you did not follow the instructions in some of the trials, what did you do instead?" "Did you have any special tricks or strategies?" and "What do you think the purpose of this experiment is?" Unlimited time was allowed for answering these questions; upon completion of the questionnaire, the purpose of the experiment was explained and questions were answered.

### Results and Discussion

The mean response times for scanning between the members of each pair of objects are plotted against the corresponding inter-object distances in Figure 1. Only correct responses were considered. The correlation between scan times and 3-D distance is high ( $r = .92$ ), and as is evident in Figure 1, times increased linearly with distance. The correlations between distance and individual subjects' response times range from .65 to .94, with a median of .78. A one-way repeated measures analysis of variance confirms that different object pairs required different amounts of time to scan,  $F(9, 81) = 8.89$ ,  $p < .001$ , and a trend analysis shows that the linear increase with distance generalizes over subjects,  $F(1, 81) = 68.44$ ,  $p < .001$ . Furthermore, the deviation from linearity is not significant,  $F(8, 81) = 1.45$ ,  $p > .10$ . Finally, the mean scan times were regressed against the distances in the two-dimensional

plane perpendicular to the line of sight while holding three-dimensional distances constant; this partial correlation is not significant ( $r = .36$ ),  $t(7) = 1.02$ ,  $p > .10$ . In contrast, the partial correlation of response times with three-dimensional distance is highly significant ( $r = .92$ ),  $t(7) = 6.27$ ,  $p < .001$ . Errors occurred in 1% of the "true" trials and did not occur more frequently for shorter inter-object distances, ruling out a possible speed–accuracy trade-off.

The results replicate and extend those of Pinker and Kosslyn (1978). Thus, the high correlation between scanning time and distance is not merely an artifact resulting from subjects' responding very quickly when they did not have a distance to scan. The slopes of the best fitting lines, which are estimates of the image scanning rate, are also similar in the two experiments: 34 msec/cm in the present experiment, 35 msec/cm for the condition in Pinker and Kosslyn (1978) employing four objects, and 37 msec/cm for the condition in that experiment employing six objects.

The purpose of the postexperimental questionnaire in the present experiment and in the others that followed was to discover whether subjects somehow deduced the purpose of the experiment and responded to implicit demand characteristics by regulating their "scan" times. If so, then the foregoing results may say nothing about how space is represented in mental images. I discarded data from any subject who claimed to use imagery in less than 60% of the trials, and temporarily discarded data from subjects who either discerned that the correlation between reaction time and distance was of interest or who confessed to using some non-imagery strategy in some percentage of the trials. If the results of the data analyses with the remaining subjects are identical to those with all subjects included, it seems reason-

<sup>2</sup> This decision task was superimposed on the scanning task for two reasons: (a) to make response time less salient to the subjects, thereby reducing the likelihood of their guessing the variables of interest in the experiment, and (b) to compare the accuracy of the responses in trials involving different distances, enabling a test for possible speed–accuracy tradeoffs. Since no such tradeoffs were evident, and since the *false* trials yielded no information concerning distance in images, response times for these trials were not analyzed.

able to conclude that subjects' guessing the hypotheses or occasionally using some non-imagery strategy is not responsible for the results. This procedure is suitably conservative, because I segregated data from subjects who guessed the time-distance relation despite (a) their unanimous and vehement assertions, when asked directly, that they did not deliberately time or otherwise alter their response and (b) the fact that the time-distance relation was only one of dozens of hypotheses offered by the subjects and thus was unlikely to have been especially salient to them. (See Kosslyn, Pinker, Smith, & Schwartz, 1979, for arguments that image scanning experiments are not contaminated by demand characteristics.)

Turning to the present results, we find that four subjects guessed that response time might be correlated with distance, and one mentioned occasionally "hearing" a tone falling in synchrony with his scanning of the image. When data from these subjects are discarded, the correlation between mean response time and distance in fact *increases* to .94; the linear trend is still significant,  $F(1, 36) = 41.18, p < .001$ , and the deviation from linearity is not significant ( $F < 1$ ).

The present results indicate that visual images preserve information about metric distance in three dimensions. They clearly eliminate theories of visual memory that posit that only topological or relational spatial information is preserved in image representations (e.g., Baylor, 1971; Minsky & Papert, 1972) and any theory that would claim that only information about the two-dimensional retinal projection of a scene is represented. However, as I have argued, it may also be incorrect to liken images to three-dimensional scale models, if images do, in fact, preserve perspective information associated with a particular vantage point.

### Experiment 2

When asked to introspect, most people report that their images of three-dimensional scenes are in fact glimpses from a definite vantage point, with some objects occluding others, distant objects appearing smaller than closer objects of the same objective size, and so on (see also Pinker, 1979). There is no need to reiterate the arguments against

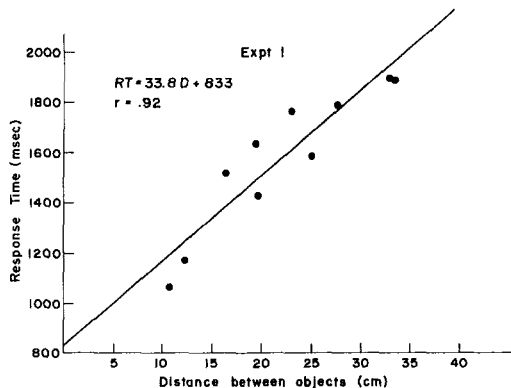


Figure 1. Mean response times for scanning mentally in three dimensions between imagined objects separated by different three-dimensional distances.

accepting introspective reports as descriptions of internal representations; needless to say, I sought to ascertain experimentally whether or not images do depict the interval distances between objects in a 2-D planar projection, reflecting the appearance of the scene as viewed from a particular direction and distance. Once again subjects were asked to scan an image, but in this case the scanning was defined in such a way that it would reflect the two-dimensional distances between objects, should these distances be preserved in images.

### Method

#### Subjects

Ten members of the Harvard community volunteered their services as subjects.

#### Materials

The visual display was similar to that used in Experiment 1, except that a toy lemon, bunch of grapes, and ball were used in place of the teddy bear, tire, and seashell. The objects' positions were chosen so that the distances between the projections of the objects' positions in the frontal plane correlated poorly with the projections onto the plane of the side of the box ( $r = .01$ ) and with the projections onto the plane of the top of the box ( $r = .08$ ). They also correlated only moderately with the distances in three-dimensional space ( $r = .56$ ). A series of trials, corresponding to the one used in the first experiment, was recorded on tape.

#### Procedure

Subjects learned to form an image of the box in the same way as their counterparts did in Experiment 1.

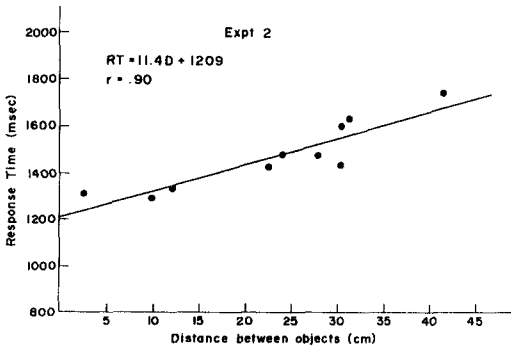


Figure 2. Mean response times for scanning mentally in two dimensions between imagined objects separated by different two-dimensional distances.

The difference between the present and previous procedure lay in the nature of the scanning instructions. Rather than focusing on a point moving from one object to another, the present subjects were to imagine that a glass plate covered the front opening of the box and that a "rifle sight" or "cross hairs" (i.e., a cross inscribed in a circle) could slide freely over its surface. When the first object was named on the tape, they were to form an image of the box and mentally "sight" the object by placing the cross hairs over it. When the second word named an object in the box, the subject was to imagine the cross hairs sliding smoothly toward that destination object until they were centered over the object, at which point the subject was to press the key labeled *true*. All other aspects of the procedure were identical to those of Experiment 1.

### Results and Discussion

The mean response times from correctly evaluated *true* trials are plotted in Figure 2. Unlike in Experiment 1 we now are examining the effects on response time of increases in the "two-dimensional" interobject distances as seen from the subjects' vantage point. These distances correlate .90 with the mean response times and from .19 to .95 with the individual subjects' response times (median = .62). An analysis of variance indicates that scanning times varied with distance,  $F(9, 81) = 6.46, p < .001$ , and a trend analysis reveals that scanning times increased linearly with distance,  $F(1, 81) = 47.61, p < .001$ , and that there was no significant deviation from that linearity,  $F(8, 81) = 1.31, p > .10$ . The partial correlation between response times and three-dimensional distances (after the shared variance with the two-dimensional distances is removed) is .37, which is not significant,  $t(7) = 1.06,$

$p > .10$ . However, the correlation between response times and two-dimensional distance with three-dimensional distance partialled out is significant ( $r = .86, t(7) = 4.43, p < .01$ ). Errors occurred in 2% of the trials and were randomly distributed across the object pairs.

Three subjects guessed that reaction times were to be correlated with distance, and one more reported occasionally "estimating" his response times. When data from these subjects are discarded, the correlation between 2-D distance and scan time arises to .92, the linear contrast remains significant,  $F(1, 45) = 32.91, p < .001$ , and the deviation from linearity remains nonsignificant ( $F < 1$ ).

The present findings are consistent with the introspections of some of the subjects in Experiment 1 that they did not feel as though they were "moving" or "flying" about in three-dimensional space, but rather that the objects in the image were always "seen" from a well-defined vantage point. The introspection that the image is "seen" as it would appear from a particular position is borne out by the fact that scan times in the present experiment accurately reflected the two-dimensional distances between objects as they appeared from the original angle of view. Thus, both the introspective reports and the data are inconsistent with Neisser and Kerr's (1973) claim that images preserve only the three-dimensional spatial layout of a scene, and not the "pictorial" or perspective properties of the retinal image.

There are two different ways in which the perspective information could have been represented in subjects' images: First, the subjects could have retained a two-dimensional "snapshot" of the original display in addition to a three-dimensional representation; alternatively, they could have generated an internal two-dimensional depiction based on the information stored in a 3-D format together with information about the original angle of view (i.e., a "vantage point" parameter). The following experiment is an attempt to discriminate between these two possibilities.

### Experiment 3

If people can use a 3-D representation to generate a 2-D depiction of the perspective

appearance of a scene from a given point of view, they should be able to "see" the planar projection of the objects from any number of viewing positions, including ones never actually experienced. That is, people should be able to study a display, imagine it rotated, say, 90°, and then "see" in their image what it looks like from the new perspective (cf. Huttenlocher & Presson, 1973; Piaget & Inhelder, 1956). This is what subjects were required to do in Experiment 3. If subjects give evidence that their images do contain information about the two-dimensional interpoint distances as seen from a novel vantage point, it seems unlikely that subjects encode just a "snapshot" or stored replica of the retinal image. In one condition of the present experiment, subjects were asked to imagine what the display looked like from the side; in a second condition, they were asked to imagine what it looked like from above.

### Method

#### Subjects

Twenty naive Harvard summer school students volunteered to be subjects in this experiment, and 10 were randomly assigned to each of the two conditions.

#### Materials

The display of objects and taped trial sequence were identical to those of Experiment 2.

#### Procedure

Subjects learned to form an image of the box in the same way as did their counterparts in Experiments 1 and 2. However, before the trials began, the experimenter covered the top and side of the box with cardboard screens and slowly rotated it 90°, asking the subject to imagine that the objects (which had in fact been removed) were rotating along with the box. In the "top" condition, the box was rotated about its horizontal axis, so that the subject was looking at its top; in the "side" condition, it was rotated about its vertical axis, so that the subject was looking at the side. The subject was then asked to "rehearse" imagining the objects through the side or top of the box. The experimenter named an object, and the subject was to say yes as soon as he or she could imagine the object in its correct position in the box as seen from the new point of view. This was repeated for the various objects until each one had been imagined four times. From then on the procedure was identical to that of Experiment 2: The subject was to imagine that a glass plate covered the side or top of the box and that cross hairs could

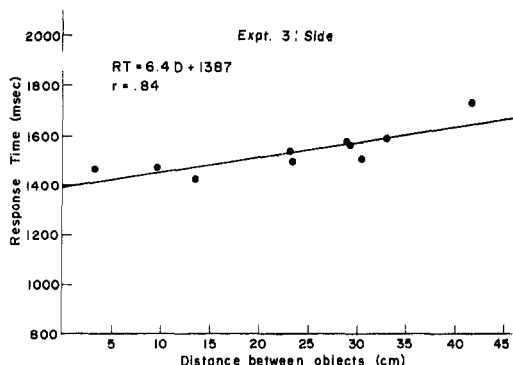


Figure 3. Mean response times for scanning mentally in two dimensions between imagined objects separated by different two-dimensional distances, following mental rotation.

slide over it and, while listening to the tape, was to "sight" the source object, scan with the cross hairs until he or she could "sight" the destination object if it was there, and then press one key if the object was in the box, another if it was not. As usual, speed-with-accuracy was stressed.

### Results and Discussion

#### Side Condition

The results of this experiment are presented in Figure 3, in which the time to scan between every possible pair of objects is plotted against the corresponding distances between the objects' projections onto the side of the box. The correlation between mean time and distance is .84; correlations between individual subjects' response times and distance range from  $-.53$  to  $.77$ , with a median of  $.52$  (two of the subjects had negative correlations). Again, times varied significantly with distance,  $F(9, 81) = 2.04$ ,  $p < .05$ , and increased linearly with distance, as shown by a significant linear trend,  $F(1, 81) = 13.01$ ,  $p < .001$ , and a nonsignificant deviation from this trend ( $F < 1$ ). One subject surmised that distance and reaction time were the variables of interest to the experimenter; removing her data from the rest leaves the linear trend significant ( $r = .81$ ),  $F(1, 72) = 9.08$ ,  $p < .005$ , and the deviation from linearity not significant ( $F < 1$ ). As expected, the means correlate poorly with the two-dimensional distances as seen from the front ( $r = .18$ ,  $p < .10$ ) and as seen from above ( $r = .43$ ,  $p < .10$ ).



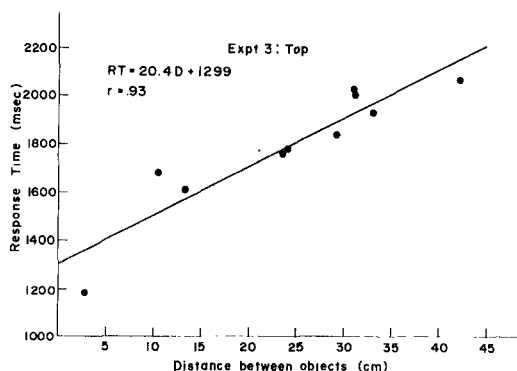


Figure 4. Mean response times for scanning mentally in two dimensions between imagined objects separated by different two-dimensional distances, following mental rotation.

However, mean response times do correlate somewhat with interobject distances measured in three dimensions ( $r = .77$ ), even when the correlation between the two- and three-dimensional distances is removed using a partial correlation ( $r = .59$ ),  $t(7) = 1.93$ ,  $p < .05$ , one-tailed. As expected, partialing out the three-dimensional distances leaves the correlation between response time and "side" distances significant ( $r = .73$ ),  $t(7) = 2.81$ ,  $p < .025$ . Errors occurred in 1% of the trials and did not occur more frequently for trials involving shorter distances.

#### Top Condition

The results of primary interest are presented in Figure 4. The mean response times correlate very highly with the distance between the objects' projections onto the top of the box ( $r = .93$ ); the response times for individual subjects correlate between .27 and .95 with distance, with a median of .58. As before, mean response times for the various object pairs differ from one another,  $F(9, 81) = 5.88$ ,  $p < .001$ , and increase linearly with distance,  $F(1, 81) = 45.46$ ,  $p < .001$ , while not deviating significantly from linearity ( $F < 1$ ). None of these effects change when data are discarded from the two subjects who guessed that response times were to be correlated with distance and from the subject who felt that he deliberately "estimated" his response times on occasion: The correlation now becomes .91, the linear trend is still significant,  $F(1, 54) = 21.06$ ,  $p < .001$ ,

and the deviation from linearity remains not significant ( $F < 1$ ). The response times do not vary significantly with any set of distances other than those seen from above: With the front view distances,  $r = .02$ ; with the side view distances,  $r = .04$ ; and with the three-dimensional distances,  $r = .50$ . In fact, the correlation between 3-D distance and scan times is now  $-.36$  when the two-dimensional top view distances are partialled out. In contrast, the correlation between response times and "top" distances remains when the three-dimensional distances are partialled out ( $r = .89$ ),  $t(7) = 5.24$ ,  $p < .01$ . Errors occurred in 1% of the "true" trials, approximately at random with respect to the different distances.

The results of this experiment then, indicate that 2-D mental image representations specific to an angle of view can be generated from an underlying three-dimensional structure, as opposed to being preserved only in a "snapshot" of the original scene.<sup>3</sup> Subjects in Experiments 2 and 3 studied the same display under the same instructions and therefore presumably encoded the same long-term representation of the display. Nevertheless, subjects appeared to have been able to construct equally accurate images whether imagining the display as it appeared from the original viewing position or as it would appear from above. However, when imagining it as it would appear from the side, the accuracy diminished somewhat, and effects of three-dimensional interpoint distances were discernible in the data. Thus I cannot rule out the possibility that the original viewing perspective has a special status as compared to new imagined perspectives.

#### Experiment 4

It has recently been shown that subjects' performance in a variety of perceptual tasks is similar to their performance in the analogous imaginal tasks (e.g., Finke & Schmidt, 1977; Moyer & Bayer, 1976; see Shepard & Podgorny, 1978, for a review). The logic involved in explaining these similarities has been spelled out by Anderson (1978). The fundamental assumption is that patterns of

<sup>3</sup> For a replication and extension of these findings, see Pinker and Finke (1980).

behavioral data observed during the performance of a cognitive task depend on how information is represented and how this representation is processed. Thus, if two tasks seem to require the same sort of process and yield similar patterns of behavior, one may tentatively conclude that they involve the same sort of representation. The recent work in imagery is a case in point, where it is argued that the representational structures underlying perception are the same as or similar to those underlying imagery. To the extent that these claims are true, progress is made in the scientific study of both phenomena, for any plausible theory of perceptual representation becomes a *prima facie* theory of imagery representation, and vice versa. Conversely, findings that constrain or falsify a theory in one domain bear directly on theories in the other.

It therefore seems important to discover whether the geometric information available to mental image processes is also available to perceptual processes. We have seen that subjects under imagery instructions do seem to access representations that display both the three-dimensional interpoint distances and the two-dimensional interpoint distances specific to a particular angle of view. In contrast, one might suppose that humans do not have access to their retinal images or to any other two-dimensional representation of the visual field during normal perception, but process a three-dimensional representation of the layout of a scene (as Attneave, 1972, and Gibson, 1966, seem to suggest). Given that we have now observed some of the characteristics of image representations, it is important to investigate whether the corresponding perceptual representations have the same properties. If they do not, it would call into question the notion that images and percepts share the same underlying representational format. Hence, I conducted a perceptual control for Experiment 2, requiring subjects to scan a scene that was visible to them at the time.

### *Method*

#### *Subjects*

Eight naive Harvard summer school students volunteered to participate as subjects; data from an additional subject were discarded after the experiment be-

cause she confessed to having followed the instructions only 45% of the time.

#### *Materials*

The display and taped trial sequence were identical to those in Experiments 2 and 3.

#### *Procedure*

Subjects were told that they were participating in an experiment on visual scanning and were asked to support their heads on the chinrest and familiarize themselves with the stimulus display. They then received scanning instructions identical to those of Experiments 2 and 3 (i.e., they were to "see" cross hairs sliding over an imaginary glass plate), except that they were to keep their eyes open and scan over the display, which was left uncovered. All other details of the procedure were identical to those of the image-scanning experiments.

### *Results and Discussion*

The mean latencies for correct responses are plotted in Figure 5. The correlation between mean response time and two-dimensional distance is .95; the correlations between individual response times and distance fall around a median of .63, ranging from  $-.16$  to  $.93$  (only one subject's correlation was negative). As before, different amounts of time were required to scan different distances,  $F(9, 81) = 3.22, p < .005$ . Furthermore, times increased linearly with increasing distance,  $F(1, 81) = 26.38, p < .001$ , and showed no other systematic variation ( $F < 1$ ). The means correlate poorly with the distances measured in three dimensions if the two-dimensional distances are partialled out ( $r = .11$ ); however, if the three-dimensional distances are partialled out, the correlation with the 2-D distance is still significant ( $r = .93$ ),  $t(7) = 6.86, p < .001$ .

One subject suspected that response times were to be correlated with distances, and another confessed to "timing" his responses on some trials, but discarding their data leaves the results unchanged: The correlation between time and distance increases to .96, the linear contrast remains significant,  $F(1, 45) = 24.43, p < .001$ , and the deviation from linearity remains nonsignificant ( $F < 1$ ). Errors occurred in 1% of the *true* trials and are uncorrelated with distance.

The results demonstrate that unpracticed subjects—and not just artists, draftsmen, and marksmen—can have access to two-

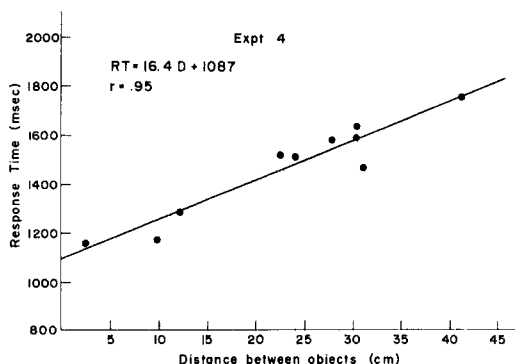


Figure 5. Mean response times for scanning in two dimensions between viewed objects separated by different two-dimensional distances.

dimensional interpoint distances during normal perception, and thus that the representations depicting two-dimensional interpoint distances during imagination can also be invoked during perception. Before this claim can be made with confidence, however, it is necessary to eliminate a possible source of artifact: the time necessary to move one's eyes from one target to another. Perhaps scanning effects in perception merely mimic those in imagery because the farther apart two objects are in two dimensions, the longer it takes to move one's eyes from one object to the other. This possibility was examined in Experiment 5.

### Experiment 5

It is not obvious *a priori* whether eye movement times should, in fact, be highly correlated with the two-dimensional distances between objects. First, it may take additional time for the eyes to accommodate and converge properly for objects at different depths, causing eye-movement times to depend on both the two- and the three-dimensional separation between objects. Second, the eyes do not always arrive precisely at their intended targets; one or more corrective movements after the initial saccade are often required. Since each such movement requires a fixed amount of time for its initiation (about 250 msec; see Fuchs, 1976), the effects of distance may be diluted or masked altogether. In fact, the structure of saccadic movements shows great variability across time and in-

dividuals and even between the two eyes of a single individual (Bahill & Stark, 1979). Third, the respective muscles that move the eyes horizontally and vertically do not begin and end in tandem but overlap to varying degrees, yielding trajectories that vary from diagonal lines to L shapes (Bahill & Stark, 1979). Thus, in this experiment I attempted to discover how the 2-D distance (visual angle) between objects affects the time it takes to move one's eyes from one object to another.

### Method

#### Subjects

Seven Harvard summer school students, one undergraduate research assistant, and two graduate students volunteered to participate as subjects.

#### Materials

The visual display and taped trial sequence were the same as those in Experiments 2, 3, and 4.

#### Procedure

Subjects were told that they were participating in an experiment on "looking." They were to listen to the tape, chin in chinrest, and were to stare at the small dot affixed to the first object mentioned in a pair on the tape. When they heard the second object named, they were to look over to it as quickly as they could if it was among those in the box and press the *true* key as soon as they were staring at the dot on the second object, and not before. If the second object was absent, they were to press the *false* key. From this point on the experiment was identical to Experiment 4.

### Results and Discussion

Figure 6 displays mean response time as a function of two-dimensional interobject distance. The mean response times correlate .89 with 2-D interobject distance; individual subjects' response times correlate from .27 to .84 (median = .56) with distance. Different amounts of time were required to scan over different distances,  $F(9, 81) = 4.40, p < .001$ , and times again increased linearly with distance,  $F(1, 81) = 31.43, p < .001$ , the deviation from this linear trend being non-significant,  $F(8, 81) = 1.02, p > .10$ . Unlike before, however, the magnitude of increase in time with distance was very small; the slope of the best fitting linear function was

only 4.4 msec/cm (5.0 msec/degree of visual angle). Three-dimensional interobject distances did not appear to contribute to response times: The partial correlation between response times and three-dimensional distances, removing the effects of two-dimensional distances, is  $-.06$ . However, removing the effects of three-dimensional distances does not eliminate the effects of two-dimensional distances; the partial correlation is  $.85$ ,  $t(7) = 4.34$ ,  $p < .01$ . Errors occurred in less than 1% of the *true* trials and were too infrequent to compare across different distances.

At first glance, the present results seem somewhat troublesome. I wish to argue that the dependence of scanning time on two-dimensional interobject distances in Experiments 2 and 4 was due to an internal scanning process operating on a mental representation common to imagery and perception. However, we see in this experiment that simply moving one's eyes takes more or less time depending on the distance in two dimensions between source and destination. It is some consolation that the speed of moving one's eyes (4.4 msec/cm) is significantly faster across subjects than the speed to scan an image in two dimensions (11.4 msec/cm),  $t(18) = 2.14$ ,  $p < .05$ , two-tailed. Nevertheless, a critic could argue that both the perceptual and the imagery findings simply reflect the amount of time taken to execute proportionally longer eye movements. After all, several theories of visual memory (e.g., Hebb, 1968; Noton & Stark, 1972) posit that images are amalgams of parts encoded during successive eye fixations on the original stimulus. In this view, the different parts are joined together by the neural commands that directed the eyes from one part of the stimulus to the other. Thus, the scanning of images in Experiment 2 could correspond to activating a neural representation of the source object (which need not occur in any sort of internal "spatial" medium), activating a trace of the neural command that drove the eyes from there to the destination object during the study phase, and then waiting until the eye movement or a trace thereof was complete before responding. Since the previous experiment showed that longer interobject distances produce lengthier eye movements,

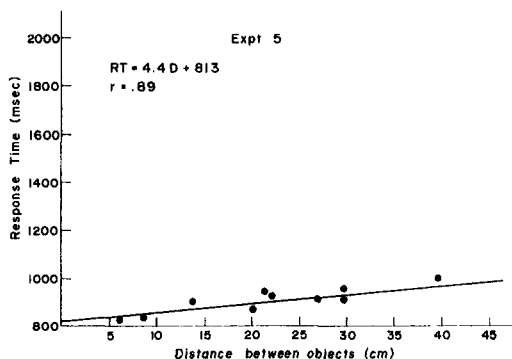


Figure 6. Mean response times for looking from one object to another when objects are separated by different two-dimensional distances. (The range of distances used corresponds to visual angles ranging from  $7^\circ$  to  $42^\circ$ .)

the linear relation observed would be a natural result.

This possible counterinterpretation of the earlier findings seems unlikely, however, for the following reasons: First, in Experiment 1, subjects scanned an image by tracing an imaginary point from one object to another, and response times depended on the three-dimensional distances between objects, which, according to Experiment 5, do not seem to influence eye-movement times. Furthermore, in Experiment 3, in which subjects scanned images representing novel views of the display, response times depended on the two-dimensional interobject distances as seen from a viewpoint the subject never actually experienced. Thus the response times could not simply have reflected the neural eye-movement commands that guided the inspection of the display during the study phase. In fact, very different results were obtained in Experiments 1, 2, and 3, although all the subjects studied the same display in the same way, unaware of the task to follow. Thus, the most parsimonious explanation for the imagery results would posit a single three-dimensional representation as the underlying basis for subjects' performance in all those tasks.

### Experiment 6

The logic of the argument against the eye-movement interpretation of Experiments 1 and 2 was used to motivate the next and

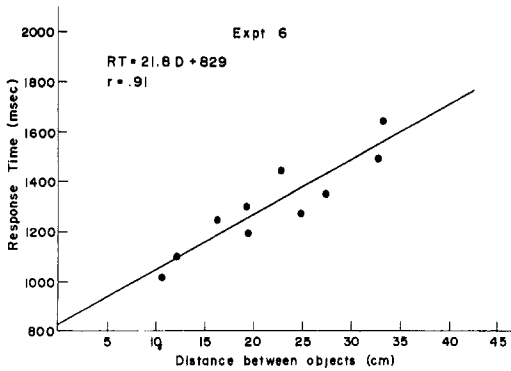


Figure 7. Mean response times for scanning in three dimensions between viewed objects separated by different three-dimensional distances.

final experiment. I have argued that "scanning an image" is not an artifact of stored eye-movement commands, because images can be scanned in depth as well as in two dimensions. In this experiment, I argue that "scanning a percept" cannot be an artifact of actual eye movements, *if* percepts can be scanned in depth as well as in two dimensions.

### Method

#### Subjects

Six Harvard undergraduates and two Boston University graduate students served as subjects. Data from two other subjects were discarded because their mean response times exceeded 4.5 sec, which was more than 3.5 times as large as the mean of the other subjects. This criterion, incidentally, does not eliminate any other subject in any other experiment reported in this article.

#### Materials

The stimulus display and taped trial sequence were identical to those of Experiment 1.

#### Procedure

The procedure was identical to that of Experiment 4, except that subjects were given the same scanning instructions as their imagery counterparts in Experiment 1. That is, they were told to look at the first object named, then wait for the tape to name a second object; if it was in the box, they were to track an imaginary small point or black dot moving smoothly in a straight line from the first object to the second and were to press the *true* key when their gaze arrived at that object. All other details were identical to those of Experiment 4.

### Results and Discussion

The mean latencies for correct responses are plotted against corresponding three-dimensional interobject distances in Figure 7. Three-dimensional distance correlates .91 with mean response times and from .43 to .90 with individual response times (median = .80). The effects of distance are significant in an analysis of variance,  $F(9, 81) = 6.11$ ,  $p < .001$ , and response times increased linearly with distance as shown by a significant linear trend,  $F(1, 81) = 45.86$ ,  $p < .001$ , and a nonsignificant deviation from linearity,  $F(8, 81) = 1.14$ ,  $p > .10$ . However, the partial correlation of the means with the two-dimensional distances, removing the effects of the three-dimensional distances, is highly significant ( $r = .87$ ),  $t(7) = 4.67$ ,  $p < .005$ . Although the nonsignificant deviation from the linear trend of 3-D distance advises against testing any further trends, it was of interest to test whether the component of two-dimensional distance that is uncorrelated with three-dimensional distance translates into a statistically significant contrast; that is, whether the unconfounded effect of 2-D distance on response time generalizes over subjects. Indeed, this orthogonal 2-D linear trend is significant,  $F(1, 81) = 6.68$ ,  $p < .025$ . In any case, the correlation of response times with three-dimensional distances remains significant after the two-dimensional distances are partialled out ( $r = .97$ ),  $t(7) = 10.14$ ,  $p < .001$ . No subject in this experiment suspected any of its purposes or reported using any special strategy. Errors occurred in less than 1% of the trials and were too infrequent to compare across different distances.

The results indicate that scanning a visual display in three dimensions is controlled primarily by the distance scanned in three-dimensional space, with a smaller influence being exerted by the distance scanned in two dimensions. It seems clear that the effect of distance on time to scan a visual display is not simply due to the time it takes to move one's eyes. Nevertheless, it would not be surprising if eye movements, whose durations reflect 2-D distance, exert an effect on response times as well, causing the significant partial correlation of time with 2-D distance

that was observed in this experiment.<sup>4</sup> Since vision may be suppressed during saccades (see Volkmann, 1976), it may be impossible to guide scanning in three dimensions while one's eyes are moving. Thus, eye movements and scanning may occur on a mutually exclusive or time-sharing basis, causing their effects on response time to add. That is, one may iteratively move one's eyes a discrete amount, scan the visible scene in three dimensions, and then move one's eyes to take in the next successive "frame." In this view, then, when eye movements and three-dimensional scanning do not compete for the same information, as when one scans a mental image in three dimensions with eyes closed, we would not expect the two-dimensional distances to influence response times, and indeed, in Experiment 1 (and in Pinker & Kosslyn, 1978; see Pinker, 1979), they did not.

### General Discussion

We have seen that once people have studied and memorized the appearance of a three-dimensional scene, they have the ability to construct and use mental images depicting that scene in a variety of ways. The existence of these abilities places constraints on possible theories of the mental representation of visual information. In this section I consider three issues in particular: the format of the representational structures underlying images of three-dimensional scenes, the integration of these structures into a general model of imagery, and the process of scanning images and percepts in three dimensions.

### *Representational Structures*

Marr (1978) and Marr and Nishihara (1978a, 1978b) have proposed that three distinct types of structures represent visual information during the recognition of three-dimensional shapes. According to Marr and Nishihara, visual information is transformed from one format to the next in the course of perception. The first and most peripheral representation, which they call "the primal sketch," is a two-dimensional array that makes explicit the intensity changes and local two-dimensional geometric properties of the retinal image. The second representation,

which they call the "2½-D sketch," represents the depths and orientations of each point on the visible surfaces of objects relative to the viewer's vantage point. This information is displayed in a coordinate system that is centered on the vantage point and hence is called a "viewer-centered" representation. The third representation, and the one that feeds into the shape recognition process, is called the "3-D sketch." In this format, objects are represented as a set of "volumetric shape primitives" organized within a three-dimensional coordinate system. This coordinate system is defined by the natural axes of the object and hence is called an "object-centered" representation. Since this is the only explicit and general model of the perception of three-dimensional objects, and since it is likely that imagery and perception share some of their representational structures, one is led to ask whether any one of these types of representation is viable in the face of the current data.

First, people can form mental images that preserve the metric three-dimensional distances between objects in a scene. Therefore, mental images can be neither simple two-dimensional snapshots of visual scenes nor like Marr and Nishihara's "primal sketch." Second, people can form images that preserve two-dimensional metric interpoint distances as they would appear from the original viewing angle. Therefore mental images are neither like simple three-dimensional scale models, in which no particular angle of view is defined, nor like Marr and Nishihara's "object-centered" representation. Third, people can form images that display two-dimensional metric interpoint distances as they would appear from a new vantage point, never actually seen. Therefore, mental images are neither like simple dioramas nor like the "2½-D sketch" of Marr and Nishihara. Our findings, then, underline the flexibility of the representational system. Whatever the underlying data structures and processes are like, taken

<sup>4</sup> In a separate experiment using a different configuration of objects and different subjects (Pinker, 1979), I again found that two- and three-dimensional distances independently predict components of the response times.

together the system is not constrained to represent objects in either a "viewer-centered" or "object-centered" manner. Rather, we seem to encode a sufficiently rich representation and have sufficiently powerful transformation operations to have access to a great many forms of information about an object's or scene's appearance. Any one of Marr and Nishihara's representations taken alone cannot account for the present findings.

### *A Model of Imagery*

Although neither a 2-, 2½-, nor 3-dimensional representation is sufficient in itself to account for performance in the present experiments, a model that would incorporate several types of interacting representations and processes has more promise. I examine how one such system, embodied in a computer simulation of two-dimensional imagery (Kosslyn & Schwartz, 1977, 1978; Kosslyn et al., 1979) could be extended to account for the representation and processing of three-dimensional visual information.

In the simulation model, the visual information embodied by images is stored in long-term memory as a file containing a list of coordinates of the points defining the encoded shape, with the origin of the coordinate system centered on the shape. Images are formed by placing the points specified by these so-called "deep" files onto a single two-dimensional "surface" array, which is centered on the viewer's "fixation point." The subroutine that performs this mapping, *PICTURE*, can depict shapes encoded in different files at various sizes, locations, and orientations in the array. Once a scene is depicted in the surface array, it is accessible to processes that interpret or transform the patterns displayed.

It is not hard to see how one could adapt this representational system so it can handle three-dimensional information. First, the deep representation must be altered so that shapes can be defined with respect to a three-dimensional instead of the current two-dimensional object-centered coordinate system. The shapes themselves could be

represented relative to this system as lists of polar coordinates ( $R, \Theta_1, \Theta_2$ ), analogous to the list of coordinates currently used ( $R, \Theta$ ); alternatively, they could be represented as a set of Marr and Nishihara's "volumetric shape primitives" ("generalized cones" of various shapes and sizes, see Marr, 1978; Marr & Nishihara, 1978a, 1978b). At present, it seems difficult to distinguish these possibilities. Second, the *PICTURE* subroutine would have to be more "intelligent," containing something like an algorithm that takes a 3-D, object-centered representation of a shape, together with a specification of the relative angle and distance of the vantage point, and computes the regions of the surface array that should be filled in to depict the perspective view of the object correctly. Finally, one might alter the surface array itself, making it resemble the 2½-D viewer-centered representation of Marr and Nishihara (whereby the array cells contain not a dot, but a vector representing the depth and surface orientation relative to the viewer of the corresponding local region of the visible surface), instead of the current simple two-dimensional viewer-centered array. This would allow the third dimension to be represented in images, while preserving perspective effects specific to a particular vantage point.

### *Image Scanning*

How would image scanning in three dimensions proceed according to this model? Kosslyn and Schwartz (1977) and Kosslyn et al. (1978) originally argued that scanning should be represented by shifting the image across the surface matrix and not by the movement of a fixation point or marker across the image itself. This mechanism was motivated in part by the introspection that it is easy to scan around the four walls of a room, never hitting the edge of one's image, and by Kosslyn, et al.'s finding that people can scan beyond the boundaries of what was represented in their image when scanning was initiated. On this account, scanning to "overflowed" regions involves a continuous process of constructing new portions of the image at the "leading edge"

of the array, and then shifting the material toward the center part of the array, which is the part with the highest resolution.

A straightforward extension of this account to model three-dimensional scanning would involve moving the scene relative to the viewer in three dimensions. One could increment the parameter representing the position of the vantage point relative to the scene, indicating a small movement of the scene relative to the viewer in any direction in 3-D space. This new value would then be fed into the 3-D-to-2½-D mapping process, which would alter the surface image so as to display the scene as it would appear following the small movement. This sequence could be executed iteratively, simulating movement of the scene relative to the viewer in three-dimensional space and causing scanning times to be proportional to distance in space between source and destination.

Unfortunately, this account runs into two problems. First, it is inconsistent with the results of Experiment 6, in which subjects scanned a visible scene in three dimensions. In that situation, surely the scene did not appear to move relative to the vantage point (barring the unlikely possibility that subjects moved a "ghost" image of the scene towards them, suppressing the visual input), yet scanning times still correlated with distance in three dimensions. Second, when subjects were questioned after the image-scanning experiment, they denied the experience of "moving" through space, or of seeing objects "loom large" or even "approach" as they scanned towards them. Rather, the scene seemed more or less stationary, and they claimed that their fixation point seemed to change relative to it.

The foregoing sort of introspective report suggests a somewhat different account. Subjects may have focused on a "fixation point" and then moved it relative to the scene in three dimensions by smoothly altering its coordinates in the 2½-D surface matrix. If the fixation point was assigned initially to the array cells occupied by the "source" object and was shifted across successive cells on an imaginary line in three-

dimensional space that led to the destination object, scanning times would reflect 3-D interobject distances. According to this view, scanning is similar to a "region-bounded" translation transformation, in which a point or set of points is shifted relative to the rest of the scene, which would remain stationary within the array. This is the possibility that Kosslyn and Schwartz and Kosslyn et al. rejected earlier, but there is no reason why this mechanism should be exclusive of or incompatible with their alternative mechanism of shifting the image pattern across the array. Assuming it is possible to scan smoothly beyond the edge or horizon of an image, one could bring in new material from the deep files as one's fixation point approached the edge or horizon. And with any particular glimpse of the scene loaded into the array, one could shift the fixation point relative to it in any direction. Presumably "bumping into the edge" could be avoided by coordinating these two processes smoothly. One would never bring new material into the array in so big a jump that the fixation point was "knocked off" the trailing edge before it could be recentered, nor would the fixation point move so quickly that it "ran off" the leading edge before new material could be brought in. This is analogous to a boy's attempting to remain on a "down" escalator indefinitely by climbing the moving steps: he must not climb so quickly that he walks off the top, nor so slowly that he is pushed off the bottom, but within these bounds he can be at any place along the escalator that he wishes.

It is interesting to note that the explanation for image scanning (such as occurred in Experiment 1) closely parallels the explanation for perceptual scanning (such as occurred in Experiment 6). In the perceptual case, I argued that one process (eye movements) brought new material into the internal visual array, and a second ("scanning") shifted an internal fixation point within that medium (cf. Kaufman & Richards, 1969; Sperling, 1960). In fact, it was possible to discern the separate effects of these two processes on the response times: Eye movements caused them to correlate with two-



dimensional interobject distances, and the addition of mental scanning caused them to correlate with three-dimensional interobject distances as well. Similarly, I posit two processes at work in image scanning: shifting the internal fixation point and bringing new material into the array. When shifting a mental image, there is nothing analogous to the visual suppression accompanying eye movements, so there is no reason to believe that scanning and shifting the image cannot be coordinated so as to proceed simultaneously. Thus one would not expect (and in fact, Experiment 1 did not show) independent effects of 2-D and 3-D distances on the data.

In conclusion, it appears that the study of the mental representation of three-dimensional visual space, though still in its infancy, is a tractable enterprise. At present, a model with the general architecture of the Kosslyn and Shwartz simulation, and with information structures of the Marr and Nishihara sort, seems to be a promising first approximation to a model of this cognitive faculty.

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